GTreasury Senior Software Engineer

Craft Exercise

Case Study

Overview

In this challenge, you'll build a RESTful API for a simplified event ticketing system.

Requirements

Core Functionality

Create a RESTful API that supports the following features:

1. **Event Management**
    a. Create, retrieve, update, and delete events
    b. Each event should have a name, description, venue, date, time, total ticket capacity, and pricing tiers
2. **Ticket Management**
    a. Purchase tickets for an event
    b. View ticket availability
    c. Handle ticket inventory and prevent overselling
3. **Basic Reporting**
    a. Generate a summary of ticket sales by event

Technical Requirements

- Implement the solution in .NET
- Use any framework, plugins, and 3rd party components that you're comfortable with
- Use a database of your choice (relational or NoSQL)
- Include proper error handling and validation
- Write meaningful tests (unit and/or integration)
- Include a README with setup instructions

AI Tooling Guidelines

- Use of AI tools is permitted (e.g. CoPilot), we ourselves have been adopting them to improve developer productivity
- If used, briefly note in your README or architecture doc how AI helped you accelerate your outcomes.
- We're generally more interested in understanding the choices you've made, tradeoffs considered, thoughts on further operationalizing, and overall thought process on the approach

Evaluation Criteria

Your solution will be evaluated based on:

1. **Code Quality**
    a. Clean, readable, and well-structured code
    b. Proper error handling and edge cases
    c. Appropriate use of design patterns
2. **System Design**
    a. API and RESTful design
    b. Data model design
    c. Scalability considerations
3. **Testing**
    a. Test coverage and quality
    b. Edge case handling
4. **Documentation**
    a. Setup instructions
    b. Design decisions and trade-offs explained

Submission Guidelines

- **Submit your code by sharing a link to a private GitHub repository with both your recruiter and recruiting coordinator by the provided deadline.**

Time Expectation

We respect your time and expect this challenge to take approximately 4-6 hours. Focus on demonstrating quality over quantity.

Questions

If you have any questions or need clarification about the requirements, please don't hesitate to reach out to the Recruitment Team.